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Individual Assignment

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1.0 Introduction

This report builds and evaluates a FinBERT news-sentiment trading strategy using one-minute price data for MI Technovation Berhad (5286) from 13 April to 8 July 2026. The financial headlines are converted into trading signals and tested across multiple holding periods under defined risk and cost assumptions. The performance is then judged based on return, risk and trading-efficiency metrics.

2.0 Data Preparation

Cleaning Check	Result
Initial Dataset Size	19,928 rows
Incomplete Records Removed	5,338 rows
Final Verified Dataset	14,590 rows
Duplicate Records Identified	0 rows

Figure 1. Price-data cleaning verification for MI Technovation Berhad.

Source: Author's calculation using LSEG Workspace data (2026).

Figure 1 shows that originally, the dataset had 19,928 timestamped observations. After having removed the 5,338 cells with missing closing price data, 14,590 timestamped observations remained and did not find any duplicate timestamps. The rows with missing closing prices were simply removed instead of estimated so that my backtest used observed prices only, since data-cleaning and aggregation decisions for high-frequency financial data can materially affect subsequent statistical results (Brownlees & Gallo, 2006).

Verification Measure	Result
Average daily return	0.8849%
Required condition	Greater than 0%
Outcome	Satisfied

Figure 2. Positive average daily-return verification for MI Technovation Berhad.

Source: Author's calculation using LSEG Workspace data (2026).

Figure 2 contains an arithmetic mean of 0.8849%; it is positive, satisfying the selection criterion laid down in the question. For the LSEG stock-selection period, the whole-period change was not used, as it would not make sense to report an endpoint return instead of the mean return of the stock, which would have to be calculated daily.

3.0 Descriptive Statistics and Data Characterization

Statistic	Result
Mean daily return	0.8849%
Median daily return	0.6452%
Minimum daily return	-6.2241%
Maximum daily return	9.9476%
Daily-return standard deviation	3.1641%
Daily-return skewness	0.2946
Daily-return kurtosis (Pearson)	3.4146
Minimum one-minute closing price	RM2.86
Maximum one-minute closing price	RM5.06
One-minute closing-price range	RM2.20
Number of trading days	58
Daily-return observations	57

Figure 3. Descriptive statistics of MI Technovation Berhad.

Source: Author's calculation using LSEG Workspace data (2026).

Figure 3 reports an average daily return of 0.8849% (median = 0.6452%), with returns ranging from -6.2241% to 9.9476%. The standard deviation is 3.1641%, which is much larger than the mean. This tells that even though the return is positive, there is also considerable volatility in the returns. Skewness = 0.2946 means the distribution is mildly positive but relatively balanced. Pearson kurtosis = 3.4146 means that the tails are slightly heavier than they would be if the returns were normally distributed, so there is some exposure to unusually large movements.

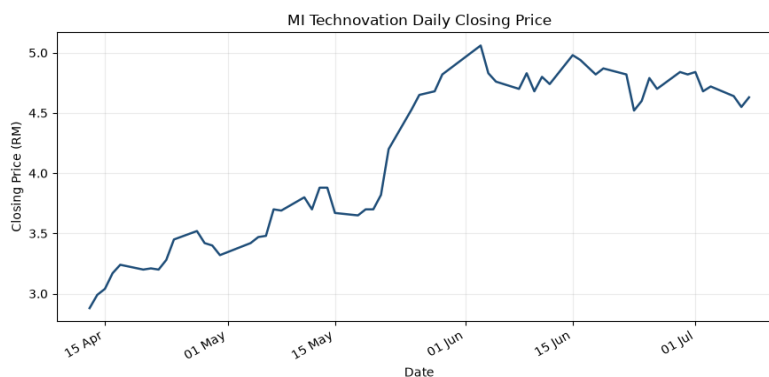


Figure 4. Daily closing price of MI Technovation Berhad.

Source: Author's calculation using LSEG Workspace data (2026).

The data used spread across 58 trading days. This generated 57 daily returns. Minute prices during these days ranged from RM2.86 up to RM5.06. This is consistent with Figure 4. Below it can see that the daily closing share price began at RM2.88 on 13 April, rose to RM5.06 on 3 June, but eventually dropped to RM4.63 on 8 July. This upward movement supports the positive mean return. The wide range of prices and the volatility in the movement give a clear justification to test various holding periods with proper risk controls in place.

Liquidity Measure	Result
Raw timestamped observations	19,928
Empty closing-price observations	5,338
Empty closing-price rate	26.79%
Valid closing-price coverage	73.21%
Valid one-minute observations	14,590
Average valid bars per trading day	251.55
Total trading volume	244,560,200
Average daily volume	4,216,555
Median daily volume	3,128,400
Unchanged consecutive valid prices	55.00%
Largest within-session interval	22 minutes
Missing volume fields remaining	0

Figure 5. Missing observations and liquidity measures.

Source: Author's calculation using LSEG Workspace data (2026).

Figure 5 shows that missing closing prices formed 26.79% of the raw observations, providing valid-price coverage of 73.21%, meaning that it had on average 251.55 valid one-minute bars each day. No empty volume fields remained. Volume traded was 244.56 million shares. Average daily volume was about 4.22 million shares compared to 3.13 million shares at median that indicates few high-volume days pulled averages well up. Furthermore, of the consecutive valid prices in the same trading session, 55.00% remained unchanged. Between consecutive observed prices, the biggest time difference was 22 minutes (10:43 a.m.–11:05 a.m.) on 5 May.

4.0 Signal, threshold and holding-period design

Forward Horizon (Days)	No. of Trades	Win Rate (%)	Average Return per Trade (%)	Cumulative Return (%)	Ending Equity (RM)	Maximum Drawdown (%)	Sharpe Ratio	Profit Factor
1	20	70.00	0.02	0.13	10012.69	−3.15	0.13	1.05
2	20	80.00	0.03	0.22	10021.96	−4.52	0.11	1.07
3	19	84.21	0.37	3.47	10347.30	−1.75	2.01	3.00
4	19	84.21	0.51	4.82	10482.37	−1.75	1.97	3.77
5	19	89.47	0.84	8.16	10815.87	0.00	2.73	42.10
10	19	89.47	0.40	3.80	10380.50	0.00	1.30	19.94
20	19	89.47	0.00	0.00	10000.00	0.00	−64.35	0.00

Forward Horizon (Days)	No. of Trades	Win Rate (%)	Average Return per Trade (%)	Cumulative Return (%)	Ending Equity (RM)	Maximum Drawdown (%)	Sharpe Ratio	Profit Factor
1	17	70.59	0.15	1.26	10126.48	−2.05	1.63	1.46
2	17	82.35	0.24	2.00	10200.05	−2.83	1.39	1.74
3	16	87.50	0.66	5.31	10531.18	0.00	3.90	27.59
4	16	87.50	0.82	6.69	10668.64	0.00	3.30	34.64
5	16	87.50	1.00	8.16	10815.87	0.00	3.01	42.10
10	16	87.50	0.47	3.80	10380.50	0.00	1.47	19.94
20	16	87.50	0.00	0.00	10000.00	0.00	−64.35	0.00

Figure 6. Backtesting results using threshold 0.0 and 0.3.

Source: Author's backtesting results (2026).

Figure 6 shows a difference in threshold levels (0.00 in the upper table and 0.30 in the lower table) for minimum sentiment values on a daily basis. By increasing the minimum value, three signals were eliminated, reducing the total entries from 20 to 17 (15%). Looking at a one-day horizon, the win rate increased from 70.00% to 70.59%, Sharpe increased from 0.13 to 1.63 and maximum drawdown dropped from −3.15% to −2.05%. With a more conservative parameter set, fewer noisy

trades enter the algorithm, therefore producing fewer but stronger signals. By making the trades more conservative, three possible trades were sacrificed when the higher minimum sentiment threshold was used. There is a fine balance between including more signals and selecting stronger ones. Since the headlines primarily capture general US market sentiment rather than specific company news, applying the more conservative sentiment threshold is considered reasonable. At the end, 0.30 was selected, although the result should be interpreted carefully considering that only 17 trades were available from the dataset.

Next, several calendar-day holding periods for up to 20 days were tested: 1, 2, 3, 4, 5, 10 and 20 days. The short-term holding periods were aimed at showing the immediate reaction of investors to information or the gradual assimilation of that information. Longer periods of 10 and 20 days tested whether the market response to the news headlines persisted or reversed. MI Technovation has daily volatility of 3.1641% and mildly heavy-tailed returns; therefore, the holding periods used seemed justifiable. The company also had intermittent liquidity throughout the day.

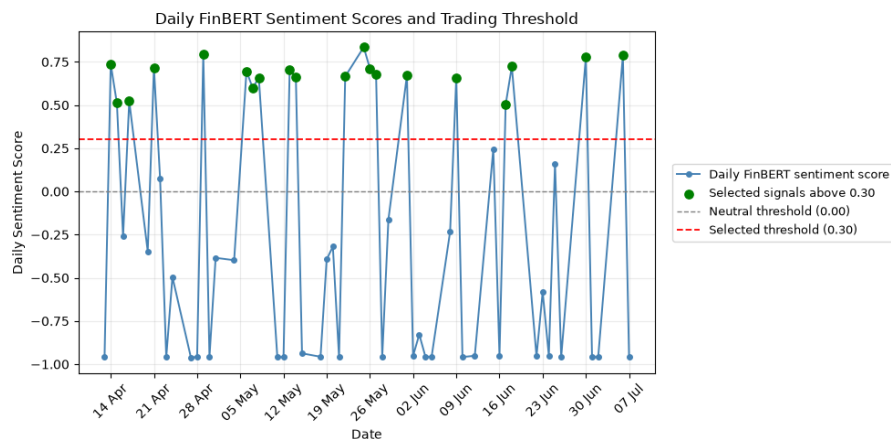


Figure 7. Daily FinBERT sentiment scores and selected trading threshold.

Source: Author's calculation using ProsusAI/finbert and the dataset (2026).

Fig. 7 shows considerable variation in the daily FinBERT sentiment during the sample period. Out of the 57 days that had headlines scored, 20 had an average sentiment score above the chosen 0.30 threshold. These observations were considered stronger positive signals. Only 17 of them were eligible for a trade to be entered because three signals had no eligible same-day price observation under the backtesting rules.

5.0 Back-Testing Assumptions and FinBERT Configuration Defense

Parameter	Final value
Minimum daily score	0.30
Holding periods	1, 2, 3, 4, 5, 10 and 20 days
Initial capital	RM10,000
Position size	50%
Round-trip transaction cost	0.38%
Annual risk-free rate	2.75%
Breakeven stop	Enabled
FinBERT model	ProsusAI/finbert
FinBERT batch size	8
Minimum headline length	12 characters
Minimum alphabetic ratio	0.55

Figure 8. Final backtesting assumptions and FinBERT configuration.

Source: Author's strategy configuration (2026).

5.1 Realistic Backtesting Assumptions

The opening balance was set at RM10,000 to represent a hypothetical retail-investor backtest account. This value was used to express portfolio outcomes in Malaysian ringgit and was given as a modelling assumption only, not intended to estimate the average account balance of Malaysian retail investors. The original position size of 100% was reduced to 50% because it was aggressive to commit all capital to each and every signal, considering MI Technovation's 3.1641% daily volatility, mildly heavy-tailed returns and intermittent liquidity. The new position size is approximately RM5,000, leaving approximately half of the account as a risk buffer.

At prices from RM2.86–RM5.06, RM5,000 purchases approximately 988–1,748 shares, or rounded share numbers of approximately 1,000–1,750, below 0.06% of the stock's median daily volume of 3.13 million shares. The assumed order should therefore have marginal or little impact on price. The original generic round-trip transaction cost of 0.50% was adjusted to 0.38%. For a RM5,000 position, the total estimated brokerage, stamp duty and clearing fees at entry and exit equals RM18.76, equivalent to 0.3752%, which was rounded to 0.38% (*Transaction Costs*, n.d.; H, n.d.).

The original default annual risk-free rate of 3.00% was adjusted to 2.75%, using Bank Negara Malaysia's Q3 2026 Overnight Policy Rate as a proxy for the risk-free rate in Malaysia (*Monetary Policy Statement – Bank Negara Malaysia*, n.d.). The breakeven stop from the original setup was maintained, as it protects the position against a reversal in price after transaction costs have been covered once the price reaches that level. However, it does not prevent a loss when the price never reaches the breakeven level.

5.2 FinBERT Configuration and Batch-Size Test

ProsusAI/finbert was retained. This is because it is trained on financial language and labels news headlines as positive, negative or neutral (Araci, 2019). Net sentiment was calculated by subtracting the negative probability from the positive probability, and the resulting value was used as the strategy's daily signal.

Batch sizes 16 and 8 were tested on the same 74 eligible headlines. When the batch size was 16, roughly five passes were required. When the batch size was 8, roughly ten passes were required. Both were executed successfully without errors. They produced identical signals and backtesting results; it was clear that batch size affected computational efficiency, not the investment conclusions. Batch size 8 was picked. Dynamic batch processing requires balancing inference latency and GPU memory usage (Zhang et al., 2024). As only a small selection of 74 financial headlines was analysed, batch size 8 was used as a cautious setting to keep inference within the capacity limits of the available computing environment.

Headlines shorter than 12 characters or with an alphabetic-character ratio below 0.55 were ignored as heuristic quality controls intended to remove very short or malformed records before sentiment analysis using FinBERT. Text preprocessing is designed to minimize noise, but its effectiveness may vary across datasets and depend on the processing choices applied (HaCohen-Kerner et al., 2020). Such settings are therefore interpreted as transparent choices for assuring data quality rather than proof of retaining every legitimate financial headline.

6.0 Performance Evaluation and Multi-Horizon Comparison

Forward Horizon (Days)	No. of Trades	Win Rate (%)	Average Return per Trade (%)	Cumulative Return (%)	Ending Equity (RM)	Maximum Drawdown (%)	Sharpe Ratio	Profit Factor
1	17	70.59	0.15	1.26	10126.48	-2.05	1.63	1.46
2	17	82.35	0.24	2.00	10200.05	-2.83	1.39	1.74
3	16	87.50	0.66	5.31	10531.18	0.00	3.90	27.59
4	16	87.50	0.82	6.69	10668.64	0.00	3.30	34.64
5	16	87.50	1.00	8.16	10815.87	0.00	3.01	42.10
10	16	87.50	0.47	3.80	10380.50	0.00	1.47	19.94
20	16	87.50	0.00	0.00	10000.00	0.00	-64.35	0.00

Figure 9. Multi-horizon performance of the final FinBERT strategy.

Source: Author's backtesting results (2026).

Figure 9 shows the entire multi-horizon result, where horizons one and two have 17 completed trades, whereas the others have only 16 because one late signal does not have enough future price data. Win rate increased from 70.59% on day one to 82.35% on day two and remained constant at 87.50% from day three onward.

Returns increased from day one to day five, where average return per trade rose from 0.15% on day one to 0.24% on day two, 0.66% on day three, 0.82% on day four and finally peaked at 1.00% on day five. Cumulative return also increased and peaked on day five, starting from 1.26% on day one and reaching 8.16% on day five, while ending equity rose from RM10,126.48 to RM10,815.87. The result started to fall back afterwards. At day ten, average return per trade was just 0.47%, cumulative return was just 3.80% and ending equity was RM10,380.50. At day 20, returns were close to zero and equity remained RM10,000, so the 87.50% win rate was not economically useful.

Maximum drawdown was -2.05% on day one and -2.83% on day two, then the backtest showed 0.00% afterwards under the breakeven-stop rule. Figure 9 shows that Sharpe peaked at 3.90 on day three before it fell to 3.01 on day five and 1.47 on day ten. Profit factor also increased from 1.46 on day one to 42.10 on day five, then declined to 19.94 on day ten. These unusually high values happened because the sample was small and there were very few negative returns. The unusual result at day 20 is further explained in Section 7.1.

At day three, it appears to have a good risk-adjusted result, with the Sharpe ratio at 3.90, but the best overall horizon is day five. At day five, average return, cumulative return, ending equity and

profit factor were all the highest. Even though the Sharpe ratio was highest at day three, it was still strong at 3.01 on day five. The results at days ten and twenty were smaller. This suggests that the sentiment effect worked mainly within three to five days and became much weaker after one or two weeks.

Hence, the five-day horizon was selected with ProsusAI/finbert, batch size 8 and the 0.30 threshold. However, it was tested in-sample with only 16–17 trades, so it is unknown whether the same result can be achieved in future trading.

7.0 Critical Reflection and Financial Interpretation

7.1 Genuine Result Anomaly

The genuine anomaly is the 20-day horizon: win rate 87.50%, average return 0.00%, cumulative return 0.00%, ending equity unchanged at RM10,000, Sharpe ratio −64.35 and profit factor 0.00. Why? The breakeven stop exited the trades at prices intended to recover the 0.38% round-trip cost, giving returns close to zero. Tiny positive values could still be counted as wins before being rounded to 0.00%. Returns had almost zero dispersion, so a small negative excess return divided by an effectively zero standard deviation produced the extreme negative Sharpe ratio. The 0.00 profit factor is similarly meaningless because the gains and losses were economically negligible. Hence, the high win rate does not mean profitability and shows why it must be interpreted together with return, risk and transaction costs.

7.2 Market Efficiency and Information Spillover

Semi-strong market efficiency states that publicly available information should be rapidly priced in (Fama, 1970). MI Technovation's results strengthened gradually. Cumulative return rose from 1.26% on day one to 8.16% on day five, while the highest Sharpe ratio occurred on day three. This indicates that the market did not immediately price in the sentiment information. It is consistent with gradual information diffusion, which may affect smaller or less-followed stocks when

information spreads slowly (Hou, 2007). Tetlock (2007) found that high media pessimism was associated with temporary downward market pressure followed by a later reversal.

MI Technovation had intermittent liquidity. Around 55% of consecutive valid prices remained unchanged, while 26.79% of raw observations contained empty closing-price fields. The longest duration between observed prices was 22 minutes. These conditions may have delayed the recorded price adjustment and allowed information to flow into more than one trading day. Despite all this, performance went down to 3.80% on day ten and approximately 0% on day 20, meaning that the information advantage eventually disappeared. This does not disprove market efficiency because the test only used around 16–17 trades across 58 trading days. It also occurred during a strong upward price trend, which may have contributed to the returns.

7.3 Transaction-Cost Drag and General Implications

Transaction costs are key factors affecting the strategy's prospects. After the 0.38% round-trip cost was included, day one resulted in an average return of only 0.15% per trade. This is a thin margin that may disappear because of bid–ask spreads, slippage or execution delays, which were not fully captured by the fixed-cost assumption. Research on momentum strategies similarly shows that transaction costs can reduce or eliminate apparently profitable returns (Lesmond et al., 2003). With the five-day strategy, the higher average return of 1.00% provides more resistance to these costs, while the breakeven stop helps prevent some profitable trades from turning into losses. However, the stop does not limit actual losses before the market reaches the breakeven level. It may also partly explain the near-zero returns at 20 days and the unusually large or unstable profit factors at longer horizons.

More generally, one-minute data and quick sentiment scoring are not a recipe for profitable intraday trading. A news-based strategy should filter out weak signals, avoid too much turnover, use realistic execution prices and costs, and align the holding period with the information life of the signal. These findings require confirmation using longer data samples, more stocks, realistic slippage modelling, non-overlapping or independently evaluated trades, and out-of-sample comparisons against buy-and-hold and no-news benchmarks.

8.0 Conclusion

The most successful application of FinBERT sentiment signals was seen in the three- to five-day settings, while the strongest overall result in terms of profitability over the trade duration was at five days. The highest average return per trade was 1.00%, with an 8.16% cumulative return and ending equity of RM10,815.87. However, the limitation of this backtest is that only 16–17 trades were made using one stock, and the in-sample period was short. Thus, more data, realistic execution costs and out-of-sample testing are needed for this strategy to be reliable in practice.

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